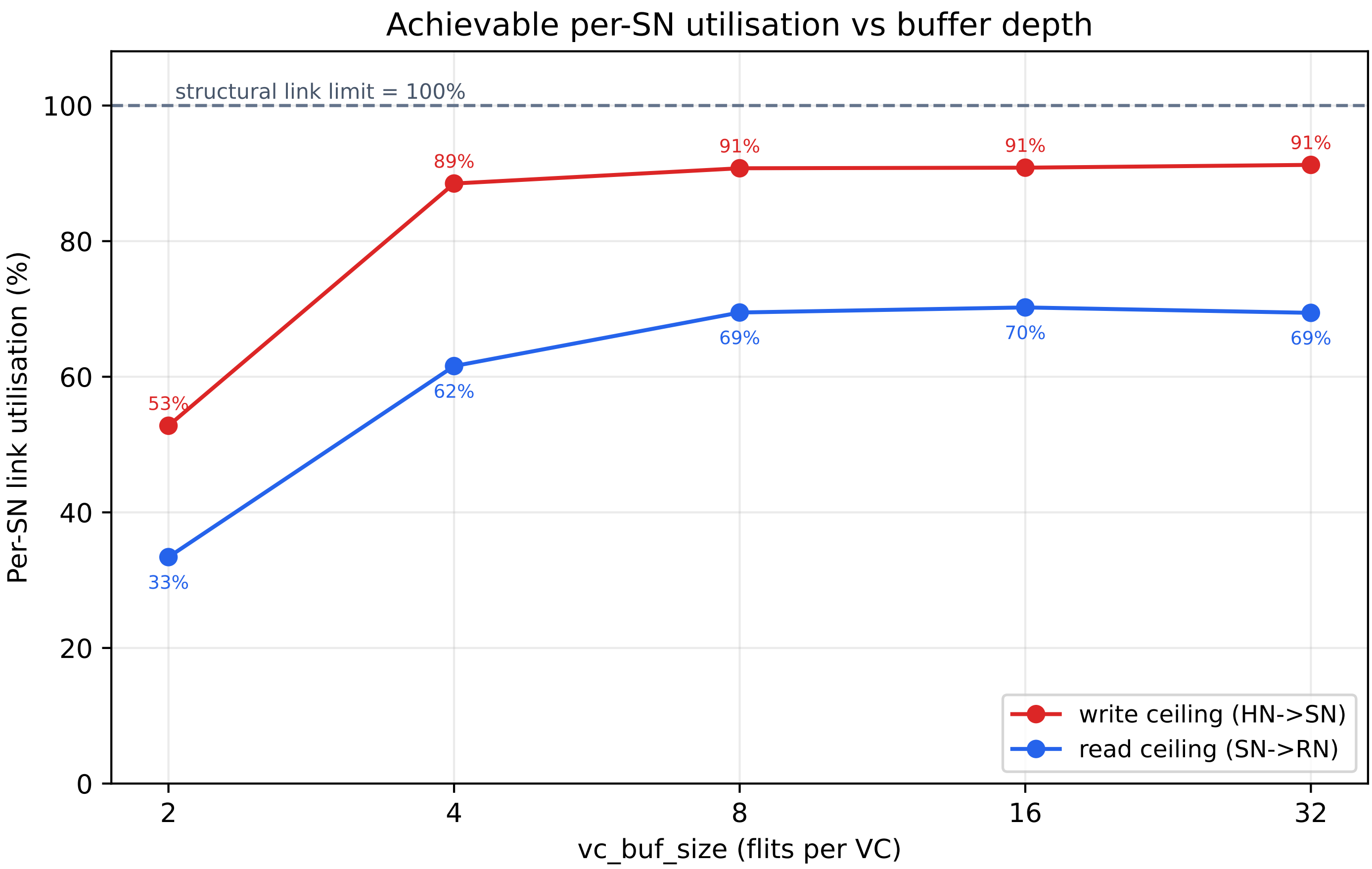
SN Throughput Ceiling vs VC Buffer Depth
6x7 mesh + 4 SN, XY routing, 2-cycle links, 2 VCs



vc_buf	read /SN (flit/cyc)	read util	write /SN (flit/cyc)	write util
2	0.334	33.4%	0.528	52.8%
4	0.616	61.6%	0.885	88.5%
8	0.695	69.5%	0.908	90.8%
16	0.702	70.2%	0.908	90.8%
32	0.694	69.4%	0.912	91.2%

Reading the curve:
- buffer 2->4 flits gives the biggest jump (read 33%->62%, write 53%->88%): the 2-flit VC buffer could not cover the link credit round-trip (2-cycle links), so the link idled waiting for credits. Deepening the buffer removes that stall.
- Write saturates near ~91% at buf>=8: the HN->SN data path becomes essentially link-bound.
- Read plateaus at ~70% even with deep buffers: the residual gap is NOT the buffer but the 84 HN -> 4 SN fan-in plus the read request round-trip (REQ into SN throttles the DMT data out). Buffer depth cannot fix a topological fan-in limit.
- Diminishing returns past buf=8: buffer is no longer the bottleneck.

Takeaway: vc_buf_size=8 captures almost all of the benefit; write becomes link-bound, read becomes fan-in/round-trip-bound. To push read further, add/spread SN nodes or shorten the coherence round-trip.

Ceiling = max sustained per-SN accepted data across the offered-load sweep, all traffic forced to the 4 SNs.